



and took a pivot into the component manufacturing business. Why the name 'Attron'? "Because it means the new wave of electronics," explains Singh.

They set up shop in Pune and started development of PMSM (permanent magnet synchronous motors). Within a year, they managed several proof-of-concept (PoC) models for two- and three-wheeler OEMs. "NDAs do not allow us to name OEMs we are doing business with, but our motors and motor controllers have helped vehicles gain 25% extra efficiency," says Singh.

The startup claims their motors offer a system efficiency of more than 97%. The average efficiency, on the other hand, is around 92%. In PMSM motors, the best efficiency being offered in the market is 82%! PMSM motors are typically used in the electric vehicle industry as they offer complete control on torque at zero speed and superfast acceleration and deceleration capabilities.

Singh explains, "It is the magnet topology that has made the difference for us. We have been awarded a patent for the same. We have

arranged magnets in a way that the magnetic linkage flux, otherwise being wasted, is being used in an efficient manner. This also allows us to use fewer magnets, resulting in reduced costs for us and for our customers."

Out of the four patents filed by Attron Automotive, three have been granted. Two are related to the stator side, and two to the rotor side. Additionally, the controllers' algorithms are developed in-house. "We have named it ETA as it represents a framework which can be customised according to customer needs," Singh says.

While the startup anticipates generating 60% of its revenue from the automotive sector, it is also exploring use cases in the agritech, aviation, defence, and consumer electronics domains. Singh notes that every segment or industry leveraging the power of propulsion and motors is an ideal business partner for them.

He adds, "Focusing on two- and three-wheelers, we will be delivering close to 3000 motors this year. The conversion duration in the automotive domain is high. At the same time, we have already started

supplying to the defence sector. We are also setting up an assembly line in Pune to increase production at the same time."

With a team of 15, Attron Automotive sources everything from India except the neodymium magnets, semiconductors, and MCUs, which are imported from China and Japan based on customer preferences for cost or quality. "If the customer wants a low-cost motor, we import the magnet from China; otherwise, we import the same from Japan or some other place," Singh elaborates.

The startup plans to utilise the recently raised ₹47.5 million funding to establish a manufacturing plant in Pune. Currently capable of producing up to 150 motors per month, they aim to increase this to 7000 to 8000 per month once the plant is operational.

"As this is our first year of operations, the paid PoCs are underway, and some orders have been received. We are in the process of finalising equipment for the manufacturing facility. Keeping control over the quality is our prime aim with self-manufacturing," Singh says.

—Mukul Yudhveer Singh

3 A CLOUD-BASED APPROACH TO BUILD BETTER BATTERIES

oorja Energy Pvt Ltd, a startup based in Bengaluru, specialises in developing software for designing, testing, and monitoring batteries post-deployment

Founded in 2022 by Vineet Dravid, Prajakta Sabnis, and Prashant Srivastava, oorja Energy merges physics and data with machine learning, simplifying the entire process of designing and testing batteries. Vineet explains that 'oorja' means energy in Hindi, making it a fitting name as their software tool aids companies in

designing better batteries.

The startup asserts its pioneering role in developing this technology. It provides a cloud-based approach, ensuring easy access to its tools for users. This is particularly important for addressing computationally intensive tasks, like simulating real-world scenarios for vehicles or stationary batteries, without

needing costly hardware. According to them, this innovative method saves time and money, differentiating it in the industry, as no other entity is currently providing hybrid (physics + ML) simulation technology.

Vineet further explains that oorja's focus is on four key aspects of battery pack design: safety, dura-